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Introduction

This document discusses a common Overlay Transport Virtualization (OTV) scenario, where the site-VLAN does not come online or the Authoritative Edge Device (AED) does not forward a set of VLANs, and the steps to troubleshoot this scenario.

Background Information

This document assumes a multi-AED setup for either unicast or multicast OTV.

Scenario

AED does not become active for a VLAN.

1. Enter the **show otv** command and check for errors. For example, 'Site-ID mismatch' is problematic.
 2. Enter the **show otv site** command and confirm the 'System-ID'.
- This is used in order to elect the AED so you know what should be active for each VLAN.

```
RTP-OTV1# show otv site
[snip]
System-ID 8478.ac0b.6044
Site-Identifier aaaa.aaaa.aaaa
Site-VLAN 6 State is Up
```

Site Information for Overlay1:

```
Local device is AED-Capable
Neighbor Edge Devices in Site: 1
```

Hostname	System-ID	Adjacency- State	Adjacency- Uptime	AED- Capable

RTP-OTV1	8478.ac0c.7b44	Full	1w6d	Yes

3. Confirm that the site-VLAN is the same on both OTV Virtual Device Contexts (VDCs) in that site.

```
Nexus-otv# show run | in "site-vlan"
otv site-vlan 7
```

The best practice is to use different site-VLANs per site.

4. Verify the Spanning Tree Protocol (STP) state of the site-VLAN along all switches in the path between OTV VDCs. It should be forwarded on the Layer 2 (L2) ports to the agg and the agg should have its ports forwarded between VDCs as well.

```
Nexus-otv# show spanning-tree vlan 7
```

```
[snip]
Interface          Role Sts Cost          Prio.Nbr Type
-----
Pol                Root FWD 1            128.4096 Network P2p
```

5. Check the event history in order to verify that hellos come in from the other AED at the local site through both the overlay and the site-VLAN and that the local AED sends two out.

```
RTP-OTV1# show otv isis internal event-history iih | in "IIH"
2015 Mar 26 14:35:09.227474 isis_otv default [14312]: [14321]:
  Send L1 LAN IIH over Overlay1 len 1397      prio6,dmac 0100.0cdf.dfdf
2015 Mar 26 14:35:08.803200 isis_otv default [14312]: [14325]:
  Receive L1 LAN IIH over Overlay1 from      SJ-OTV1 (8478.ac0c.7b47) len 1397
prio 7
2015 Mar 26 14:35:08.434157 isis_otv default [14312]: [14325]:
  Receive L1 LAN IIH over site-vlan from      RTP-OTV1 (8478.ac0c.7b44) len 1497
prio 7
2015 Mar 26 14:35:08.345369 isis_otv default [14312]:
[14321]: Send L1 LAN IIH over site-vlan len 1497      prio6,dmac 0100.0cdf.dfdf
```



Note: If the site VLAN is the same on both sites and hello packets are switched from one site to another, then OTV detects the different site ID and inactivates the VLANs in order to shut down one site. This is completed in order to avoid an L2 loop. Enter the **show otv output** command for verification. A 'Site-ID mismatch' should be displayed when this occurs. You can change the site-VLAN or stop the loop before the site-VLAN in order to correct the issue. AED can then be brought up.

6. Check the statistics in order to verify whether the hellos are received in the counters.

```
RTP-OTV1# show otv isis site statistics | begin PDU
OTV-IS-IS PDU statistics for site-vlan:
```

PDU	Received	Sent	RcvAuthErr	OtherRcvErr	ReTransmit
LAN-IIH	91697	91700	0	0	n/a

<<<<<<

```
RTP-OTV2# show otv isis traffic
```

```
OTV-IS-IS process: default
```

```
VPN: Overlay1
```

```
OTV-IS-IS Traffic:
```

PDU	Received	Sent	RcvAuthErr	OtherRcvErr	ReTransmit
LAN-IIH	85530	23298	0	0	n/a

<<<<<<

7. Check the ordinal numbers in order to ensure they are assigned correctly:

```
//////////////////////////////////// Note //////////////////////////////////////
Lower IS-IS System-ID (Ordinal 0) = AED for EVEN VLANs
Higher IS-IS System-ID (Ordinal 1) = AED for ODD VLANs
```

```
RTP-OTV1# show otv internal site
Group:239.1.1.1 Queue-size:2 num_site_adj:2 fwd_rdy_cnt:2 local-present:1
  System-ID           Up Time      Ordinal    Fwd-Rdy
  8478.ac0b.6044       1w6d         0          1
  * 8478.ac0c.7b44     1w6d         1          1 <<<<< active for ordinal 1,
  so this VDC should be AED for odd VLANs
```

```
RTP-OTV2# show otv internal site
Group:239.1.1.1 Queue-size:2 num_site_adj:2 fwd_rdy_cnt:0 local-present:1
  System-ID           Up Time      Ordinal   Fwd-Rdy
  * 8478.ac0b.6044    1w6d         0         1 <<<< active for ordinal 0,
  so this VDC should be AED for even VLANs
  8478.ac0c.7b44      1w6d         0         1
```

OTV Extended VLANs and Edge Device State Information (* - AED)

(NA) - Non AED, (VD) - Vlan Disabled, (OD) - Overlay Down
(DH) - Delete Holddown, (HW) - HW: State Down
(NFC) - Not Forward Capable

VLAN	Auth. Edge Device	Vlan State	Overlay
190	RTP-OTV2	inactive(NA)	Overlay1
191*	RTP-OTV1	active	Overlay1

<<< This AED is active for odd vlans as expected

 Note: Keep Cisco bug ID CSCur30401 in mind when you look at Ordinals!

Related Cisco Bug IDs

- Cisco bug ID CSCum54509 - Release 6.2(6), site adjacency stuck in partial state because F Series linecard packets destined to 0100.0cdf.dfdf are not flooded to Bridge Domain (BD)
- Cisco bug ID CSCur30401 - Ordinal in 'show otv site detail' is 0 for all switches

Revision History

Revision	Publish Date	Comments
1.0	22-Apr-2015	Initial Release

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